

Data profiling summary

1. Purpose

The purpose of Day 3 was to understand the structure of the Olist dataset, check data quality, identify missing values and duplicates, and find issues that need to be cleaned before further analysis.

2. Key Findings

Customers

- The Customers table has 99,441 records and no missing values.
- There are 96,096 unique customers.
- 2,997 customers made more than one purchase.
- Some postcode prefixes lost their leading zero because they were stored as numbers. These should be converted to 5 digit text values.

Orders

- The Orders table has 99,441 orders.
- Most orders were delivered successfully.
- 6,534 delivered orders were late based on the actual delivery date compared with the estimated delivery date.
- The corrected late delivery rate is 6.77%.
- The median late delivery was 7 days.
- Some orders have missing or inconsistent delivery dates and should be flagged during cleaning.

Order Items

- The Order Items table has 112,650 item rows.
- There are 98,666 unique orders in this table.
- 775 orders from the Orders table do not have a matching item record. Most of these are cancelled or unavailable orders.
- 1,278 orders contain products from more than one seller.
- 4,124 item rows have freight cost higher than the product price.
- Some very high prices and freight values were found, but they are not automatically errors.

Products

- The Products table has 32,951 products.
- 610 products are missing category, name length, description length and photo quantity information.
- These missing-category products appear in 1,603 item sales.
- 4 products have a recorded weight of 0 grams.
- There are 73 product categories.

Payments

- The Payments table has 103,886 payment records.
- One delivered order does not have a matching payment record.
- Credit card is the most common payment method.
- 80 orders have payment sequences that start at 2 instead of 1.
- There are 9 zero-value payment records.
- The largest payment value was 13,664.08, and it was confirmed to match the value of the items and freight in that order.

Reviews

- The Reviews table has 99,224 review records.
- Some orders have more than one genuine review.
- These reviews have different review IDs, so they should not automatically be treated as duplicate rows.
- 5-star reviews are the most common.
- Approximately 77% of review records are positive ratings of 4 or 5 stars.
- Many customers gave a rating without writing a review comment, which is normal.

Sellers

- The Sellers table has 3,095 sellers.
- There are no missing values and every seller ID is unique.
- Seller postcode prefixes should be stored as 5-digit text values.

Geolocation

- The Geolocation table has 1,000,163 rows but only 19,015 unique postcode prefixes.
- There are 261,831 exact duplicate rows.
- 8 postcode prefixes are linked to more than one state.
- 7 seller records and 278 customer records do not have a matching postcode in the Geolocation table.
- The raw Geolocation table should not be joined directly with Customers or Sellers because one postcode can appear many times and may multiply rows.

Category Translation

- The Category Translation table has 71 translation records.
- The Products table has 73 product categories.
- Two product categories do not have an English translation: pc_gamer portateis_cozinha_e_preparadores_de_alimentos

3. Main Issues for Day 4 Cleaning

The main cleaning tasks for Day 4 will be:

- standardise postcode prefixes as 5-digit text values;
- remove exact duplicate rows from the Geolocation table;
- prepare one usable geolocation record per postcode before joining tables;
- handle missing product information;
- flag zero product weights;
- handle payment sequence issues;
- create a clear rule for multiple reviews per order;
- handle the two missing category translations;
- keep a cleaning log explaining every change made.

4. Conclusion

Day 3 data profiling is complete. The dataset is generally suitable for analysis, but several data-quality issues need to be cleaned and validated before SQL analysis, modelling and dashboard development.